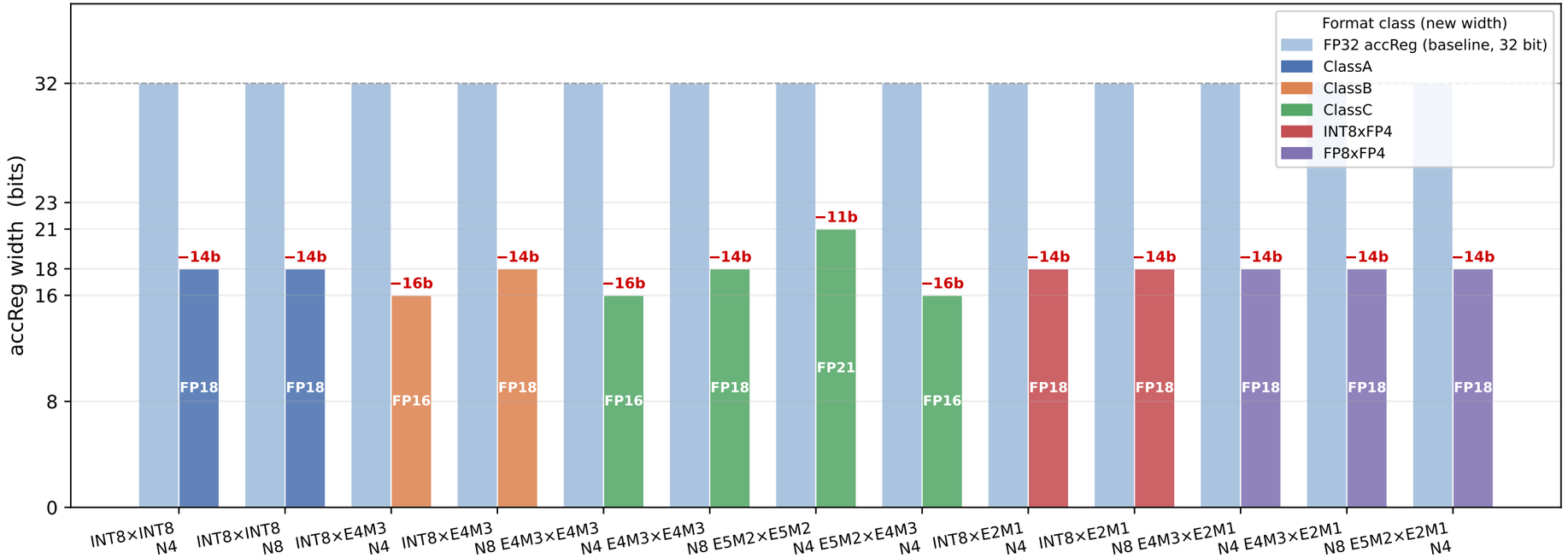
Fig 3 — Accumulator register bit-width: FP32 baseline vs. design-time parameterised
(worst-case K ∈ {4, 8, 16, 32, 64}; normErr criterion)



Note: red annotations show mantissa bits saved relative to FP32. Total reg = 1 (sign) + 8 (exp) + P (mantissa). E5M2×E5M2 requires the most bits (FP21) due to productExpRange=58; all others fit in FP16–FP18. Savings translate directly to register file area and routing reduction in the accumulation datapath.*